
Explicit vs. Implicit World Models for Continuous Control: A Lightweight Comparison on DMControl

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Abstract

World models for decision making can be divided into two broad classes: explicit models that learn latent dynamics for imagined rollouts, and implicit models that organize latent space primarily around control and planning. This paper compares these two paradigms in a lightweight continuous-control setting using a Dreamer-style recurrent state-space model and a TD-MPC2-style latent planning agent. Both agents are trained from low-dimensional state observations with saved checkpoints and reproducible logging, and are evaluated on DMControl benchmarks with cartpole-swingup as the main matched comparison and walker-walk as a lower-budget extension. Beyond return curves, the study also examines latent representations in order to assess whether explicit and implicit training objectives lead to different task-relevant structure. The results indicate that both methods learn effectively in small-scale control, while differing in optimization behavior, computational cost, and representational emphasis.

1. Introduction

World models aim to compress experience into internal representations that support prediction, planning, and decision making. Since the early “World Models” formulation of learning compact latent dynamics for control (Ha & Schmidhuber, 2018), this line of research has developed into two increasingly distinct philosophies. The first learns an explicit predictive model of latent future trajectories and uses imagined rollouts to improve behavior, as in recurrent state-space model approaches such as Dreamer and Dream-

erV3 (Hafner et al., 2020; 2023). The second learns an implicit control-oriented latent space shaped primarily by value-driven objectives, using the learned representation for planning without requiring explicit reconstruction of future observations, as in TD-MPC2 (Hansen et al., 2024). Understanding the practical trade-off between these two philosophies is the central question of this work.

The distinction is important because the two approaches place different pressures on representation learning. In explicit world models, the latent state must support multi-step prediction and reward estimation well enough for imagined trajectories to be useful for policy optimization (Hafner et al., 2020; 2023). In implicit world models, the latent state is instead shaped to be directly useful for action selection and value estimation, often through consistency objectives and temporal-difference learning (Hansen et al., 2024). As a result, the two methods may differ not only in sample efficiency and final control performance, but also in what information they preserve in their latent spaces. This makes continuous-control benchmarks such as those provided by DMControl (Tunyasuvunakool et al., 2020) a useful setting for studying both behavioral and representational differences under matched interaction budgets.

In this work, we compare these two paradigms in a lightweight state-based regime. We implement an explicit Dreamer-style agent based on an RSSM with actor-critic updates over imagined rollouts (Hafner et al., 2020; 2023), and an implicit TD-MPC2-style agent with latent dynamics, consistency learning, temporal-difference value estimation, and latent-space planning (Hansen et al., 2024). We evaluate the agents on shared continuous-control tasks and compare their learning curves, optimization behavior, and final returns. Beyond raw performance, we also investigate whether the two training objectives induce different latent geometries and different levels of task-relevant structure, since reward curves alone may not fully explain representational differences between explicit and implicit world models.

The main question guiding this paper is therefore: *under a lightweight continuous-control setup, how do explicit and implicit world models differ in learning dynamics, computational behavior, and latent representation structure?* Our working hypothesis is that explicit models will pro-

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duce more directly predictive latent states because they are trained to support imagination, whereas implicit models will produce latents more tightly aligned with control and value estimation because they are optimized around decision quality rather than explicit future reconstruction.

2. Background and Related Work

World models aim to learn compact internal representations of environment dynamics that can support downstream decision making. A foundational formulation of this idea was presented by Ha and Schmidhuber in *World Models* (Ha & Schmidhuber, 2018), where a latent model was used to compress observations and predict future behavior in a lower-dimensional space. Although the original formulation was not yet the modern standard for model-based control, it established the core intuition that control can be improved by learning a compact latent representation of the environment rather than acting directly on raw observations.

A major line of subsequent work developed *explicit* world models, in which the latent state is trained to support multi-step prediction and imagination. *Dream to Control* (Hafner et al., 2020) introduced the recurrent state-space model (RSSM), combining deterministic recurrent dynamics with stochastic latent variables in order to model temporal uncertainty while preserving memory over long horizons. The key idea is that behavior learning can be moved into latent imagination: instead of training an agent only from real environment interaction, the policy and value function are improved using imagined trajectories generated by the learned model (Hafner et al., 2020). DreamerV3 (Hafner et al., 2023) extends this paradigm into a more robust and scalable recipe across diverse domains, showing that a common world-model framework can work effectively over a wide range of tasks with relatively little task-specific retuning. In these methods, the latent state is explicitly optimized to be predictive enough that imagined rollouts are useful for policy optimization.

A second line of work has moved toward *implicit* world models, where the latent representation is shaped less by explicit future reconstruction and more by control-oriented objectives. TD-MPC2 (Hansen et al., 2024) is a representative example of this approach. Instead of emphasizing explicit observation prediction, TD-MPC2 learns latent dynamics through consistency objectives and temporal-difference learning, while planning directly in latent space (Hansen et al., 2024). This produces a representation that is tightly coupled to reward and value estimation, and therefore potentially more directly useful for action selection. In contrast to Dreamer-style methods, where imagination is central and explicit, TD-MPC2 emphasizes decision quality in latent space without requiring the same emphasis on explicit generative modeling of future observations.

This distinction between explicit and implicit latent modeling is the central conceptual axis of this work. Explicit methods, such as Dreamer (Hafner et al., 2020; 2023), place strong pressure on the representation to preserve predictive structure over imagined trajectories. Implicit methods, such as TD-MPC2 (Hansen et al., 2024), place stronger pressure on the representation to preserve control-relevant structure for value estimation and planning. These different objectives suggest that the two approaches may differ not only in reward curves and sample efficiency, but also in the geometry and semantic content of their learned latent spaces.

A broader perspective comes from recent Vision-Language-Action (VLA) models such as SmolVLA (Cadene et al., 2025) and OpenVLA (Kim et al., 2024), which highlight how action-capable representations can emerge in large multimodal systems. Although these models are not the main focus of the core comparison here, they help frame latent-space analysis as a general representation-learning question: what kind of structure is preserved when a model is optimized primarily for prediction, for control, or for action-conditioned multimodal grounding? This perspective motivates studying world-model latents not only as internal states for reinforcement learning, but also as learned representations with potentially different inductive biases.

Taken together, these works motivate the comparison developed in the rest of the paper. The goal is not only to compare two algorithms, but to compare two philosophies of representation learning for control: one that learns to imagine future trajectories explicitly, and one that learns a latent space organized around control and planning. The experiments and latent-space analysis are designed to test how these differences appear in practice in a lightweight continuous-control regime.

3. Methods

We study two contrasting approaches to world modeling for continuous control under a lightweight state-based regime: an *explicit* latent imagination agent based on the Dreamer family (Hafner et al., 2020; 2023), and an *implicit* planning agent based on TD-MPC2 (Hansen et al., 2024). Both methods operate on low-dimensional simulator states rather than pixels, which reduces computational cost and focuses the comparison on latent modeling and control rather than visual representation learning. This design is consistent with the lightweight regime considered in the study and with the use of continuous-control benchmarks under limited compute.

3.1. Explicit world model: Dreamer-style RSSM agent

Our explicit agent follows the Dreamer formulation built around a recurrent state-space model (RSSM) (Hafner et al., 2020; 2023). The model maintains a latent state composed of a deterministic recurrent component and a stochastic component, enabling it to summarize past observations while preserving temporal uncertainty. Given sequences of state observations and actions, the RSSM is trained to model latent transitions and rewards, and the learned latent dynamics are used to generate imagined trajectories for behavior learning (Hafner et al., 2020). Policy and value optimization are therefore performed on imagined rollouts rather than only on real environment interaction, which is the defining feature of the Dreamer family (Hafner et al., 2020; 2023).

In our implementation, the Dreamer agent is trained on state observations only, without an image encoder or decoder. For the main experiments, we use a GPU-based JAX setup with four parallel environments, a training ratio of 8, and periodic checkpointing. The resulting cartpole configuration contains 637,446 trainable parameters, as reported by the training script at startup. Scalar metrics are written to `metrics.jsonl`, episode returns to `scores.jsonl`, and checkpoints are saved in a dedicated `ckpt/` directory, enabling both learning-curve analysis and later latent extraction from saved checkpoints.

3.2. Implicit world model: TD-MPC2-style agent

Our implicit agent follows the TD-MPC2 paradigm (Hansen et al., 2024). Instead of explicitly training a model for latent imagination in the Dreamer sense, TD-MPC2 learns a latent dynamics model shaped by a consistency objective together with temporal-difference value learning, and performs action selection by planning directly in latent space (Hansen et al., 2024). The learned representation is therefore optimized to support control and value estimation, rather than explicit reconstruction of future observations.

As with the Dreamer agent, we use state observations only. For the main cartpole comparison, we adopt the official single-task default configuration size for TD-MPC2, namely `model_size=5`, which corresponds to approximately 4,932,704 trainable parameters in our run. Training is performed in the provided Docker environment with GPU acceleration, evaluation rewards are logged to `eval.csv`, and the final model is saved as `models/final.pt`.

For experiments beyond the main cartpole benchmark, we reduce the TD-MPC2 configuration to `model_size=1` in order to keep the study feasible under limited compute. This change is a practical engineering choice rather than a conceptual change to the method, and it is reported explicitly so that any comparison beyond the main matched benchmark is interpreted with the corresponding difference in model

capacity in mind.

3.3. Experimental protocol and fairness

The comparison is organized around shared tasks, observation modality, interaction budgets, and seed protocol. The main benchmark is `cartpole-swingup`, for which both agents are evaluated over three random seeds. The intended comparison budget on this task was 200,000 environment steps per seed, which lies within the commonly used lightweight range for DMControl experiments in the present study.

In the finalized TD-MPC2 cartpole runs, however, the three seeds ended at 400k, 200k, and 150k environment steps, respectively. The shorter stopping point for the third seed was a practical decision under time and compute constraints after performance had largely stabilized, while the longer runs were retained for exploratory inspection. To preserve fairness, all multi-seed TD-MPC2 aggregates were computed only on the shared 150k horizon, and all direct Dreamer vs. TD-MPC2 comparisons were truncated to the same range.

To broaden the empirical scope beyond a single environment, we additionally consider `walker-walk`. In this extension, Dreamer is kept in the same lightweight state-based regime, while TD-MPC2 uses `model_size=1` for computational efficiency. Unlike the cartpole setting, this additional environment is not intended as the strongest like-for-like comparison, but rather as a lower-budget extension used to test whether the qualitative trends persist beyond the main benchmark. The walker-walk experiments use a budget of 100k environment steps per seed.

Performance is measured as return versus environment interaction steps. Because the two codebases expose different local logging pipelines, the explicit agent is analyzed through episode-score and loss logs (`scores.jsonl`, `metrics.jsonl`), whereas the implicit agent is analyzed primarily through its evaluation reward log (`eval.csv`). For direct overlays, curves are aligned on the same step range. All runs are associated with fixed seeds, explicit run directories, and saved checkpoints or final models, allowing reported figures to be regenerated from saved logs. This is important for reproducibility and matches the requirement that figures and reported values be recoverable from code and saved artifacts.

4. Experimental Setup

We evaluate both agents on DMControl tasks using state-based observations only, which keeps the comparison focused on latent modeling and control rather than visual representation learning. The main matched benchmark is `cartpole-swingup`, which is the task for which the comparison is most complete and the one on which TD-

MPC2 is evaluated under its stronger single-task configuration, namely `model_size=5`.

Each cartpole configuration is evaluated with three random seeds. The intended cartpole budget was 200,000 environment steps per seed. In the finalized TD-MPC2 cartpole runs, however, the three seeds ended at 400k, 200k, and 150k environment steps, respectively. The shorter stopping point for the third seed was a practical decision under time constraints after performance had already plateaued. To preserve fairness, all multi-seed TD-MPC2 aggregates were computed only on the shared 150k horizon, and all direct Dreamer vs. TD-MPC2 comparisons were truncated to the same range.

To extend the study beyond the main matched benchmark, we additionally consider `walker-walk`. In this extension, Dreamer is kept in the same lightweight state-based regime, while TD-MPC2 is reduced to `model_size=1` for computational efficiency. The walker-walk experiments use a budget of 100k environment steps per seed. As a result, cartpole-swingup provides the strongest like-for-like comparison in the paper, while walker-walk broadens the empirical picture under a reduced implicit-model capacity and a smaller interaction budget.

Performance is measured as return versus environment interaction steps. Because the two implementations expose different native logging pipelines, the explicit agent is analyzed through saved episode-score and training-metric logs (`scores.jsonl` and `metrics.jsonl`), whereas the implicit agent is analyzed primarily through its saved evaluation log (`eval.csv`). For direct visual comparison, all curves are plotted against the same step axis, and multi-seed results are aggregated over the shared budget.

All experiments save checkpoints or final model artifacts, and each run is associated with an explicit seed-specific output directory. Figures are generated directly from saved logs rather than from manual inspection of terminal output. This ensures that the reported quantitative results can be regenerated from code and saved artifacts.

Table 1 summarizes the experiment matrix. Cartpole-swingup serves as the main matched comparison, while walker-walk broadens the empirical analysis under a reduced TD-MPC2 model capacity and a smaller interaction budget.

In both tasks, Dreamer uses the same lightweight state-based setup described above; only the TD-MPC2 model size changes between the main matched benchmark and the additional walker-walk extension.

Table 1. Experiment matrix. Dreamer uses the same lightweight state-based setup in both tasks; only the TD-MPC2 configuration changes.

Task	Budget	TD-MPC2
Cartpole	shared 150k	size 5
Walker-walk	100k	size 1

5. Results

5.1. Main matched benchmark: cartpole-swingup

The main matched benchmark in this study is `cartpole-swingup`, which serves as the clearest point of comparison between the two approaches. It is the task for which both agents were analyzed most completely across multiple seeds, and it is also the setting in which TD-MPC2 was evaluated under its stronger single-task configuration. For fairness, all cross-seed and cross-method cartpole comparisons are reported on the shared 150k horizon, since the finalized TD-MPC2 seed runs ended at 400k, 200k, and 150k environment steps, respectively. The longer runs were therefore truncated to the common horizon for aggregate analysis.

Figure 1 shows the main comparison between Dreamer and TD-MPC2 on this shared budget. Both methods are capable of learning useful behavior on `cartpole-swingup`, but their outcomes differ substantially in magnitude. At 150k environment steps, Dreamer reaches a final mean score of 115.30, whereas TD-MPC2 reaches a final mean evaluation reward of 682.05. The best mean value on the shared grid is 115.30 for Dreamer and 775.74 for TD-MPC2. These values indicate a strong empirical advantage for TD-MPC2 on this benchmark under the present experimental setup.

At the same time, the two agents differ not only in average performance, but also in how consistently that performance is obtained. Dreamer’s curve rises more gradually and remains in a lower-return regime throughout training, while TD-MPC2 improves rapidly and stays in a much stronger regime over most of the shared horizon. Thus, in terms of raw control performance on the main matched task, the implicit latent planning approach clearly outperforms the explicit Dreamer-style agent in this experiment.

5.2. Cross-seed stability

The confidence-interval plots reveal an important second result: higher performance does not imply higher stability. Figure 2 shows the Dreamer three-seed mean and 95% confidence interval, while Figure 3 shows the corresponding TD-MPC2 result on the shared 150k horizon. Dreamer exhibits a noticeably smaller uncertainty band at the end of training, with a 95% CI half-width of 33.28 at 150k, whereas TD-MPC2 has a much larger half-width of 386.47 at the

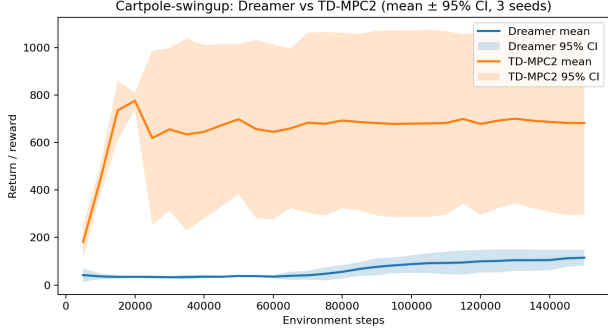


Figure 1. Cartpole-swingup comparison on the shared 150k horizon. Curves show the mean over three seeds and the shaded regions show 95% confidence intervals. The shared horizon is used because the finalized TD-MPC2 seeds did not all terminate at the same step count.

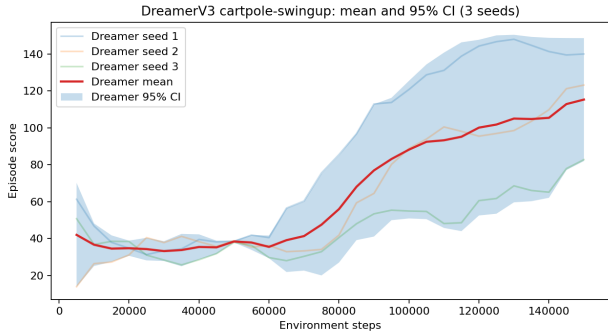


Figure 2. Dreamer on cartpole-swingup: mean learning curve over three seeds with a 95% confidence interval.

same step.

This large difference in uncertainty is consistent with the individual seed behavior visible in the per-method plots. For Dreamer, the three seeds follow different trajectories but remain within the same broad performance regime. In contrast, TD-MPC2 displays a much more uneven seed profile: two seeds converge to a strong high-reward regime, whereas one seed remains substantially worse. As a result, the TD-MPC2 mean is still much higher than Dreamer’s, but the variance across seeds is also much larger. This means that a single-seed comparison would have overstated the consistency of TD-MPC2 on this task.

The cartpole results therefore support a nuanced conclusion. On the one hand, TD-MPC2 is clearly stronger in average performance. On the other hand, Dreamer appears more stable across runs in this lightweight setup. This distinction is one of the main empirical findings of the paper: the explicit and implicit world-model paradigms differ not only in final

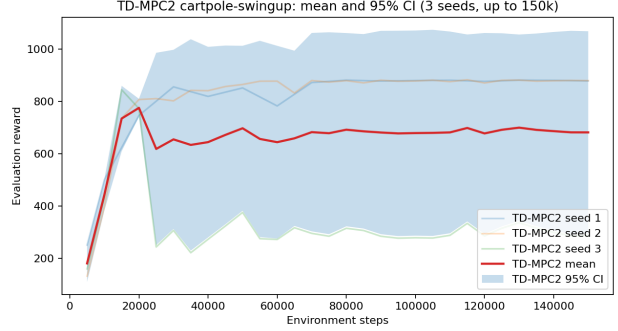


Figure 3. TD-MPC2 on cartpole-swingup: mean learning curve over three seeds with a 95% confidence interval, reported on the shared 150k horizon.

return, but also in their sensitivity to random initialization and training trajectory.

5.3. Latent-space analysis on the main task

To go beyond reward curves, we analyzed the learned latent spaces on `cartpole-swingup` using both PCA visualization and linear probes. Figure 4 presents a 2×2 PCA panel: Dreamer latents colored by reward and pole angle, and TD-MPC2 latents colored by the same quantities.

The visual difference between the two latent spaces is striking. Dreamer produces a broad, diffuse, and approximately circular latent geometry, in which both reward and pole angle vary over a large region of the embedding. In contrast, TD-MPC2 produces a thin and highly structured low-dimensional manifold, along which both reward and pole angle change smoothly. Qualitatively, this suggests that the Dreamer latent is more distributed and less directly organized around simple task-state structure, while the TD-MPC2 latent is much more tightly aligned with the progression of successful control trajectories.

This visual impression is supported by the linear-probe results summarized in Table 2. For Dreamer, a linear probe from latent state to reward achieved $R^2 = 0.7666$ with $\text{MAE} = 0.0341$. The corresponding probes for pole angle and cart position achieved $R^2 = 0.4715$ and $R^2 = 0.7993$, respectively. For TD-MPC2, the results were substantially stronger: $R^2 = 1.0000$ for reward, $R^2 = 0.9429$ for pole angle, and $R^2 = 1.0000$ for cart position, all with extremely small absolute errors.

These probe results indicate that, on the extracted cartpole trajectories used here, TD-MPC2’s latent state is much more directly and linearly predictive of both reward and key state variables than Dreamer’s latent state. In other words, TD-MPC2 appears to organize its representation much more

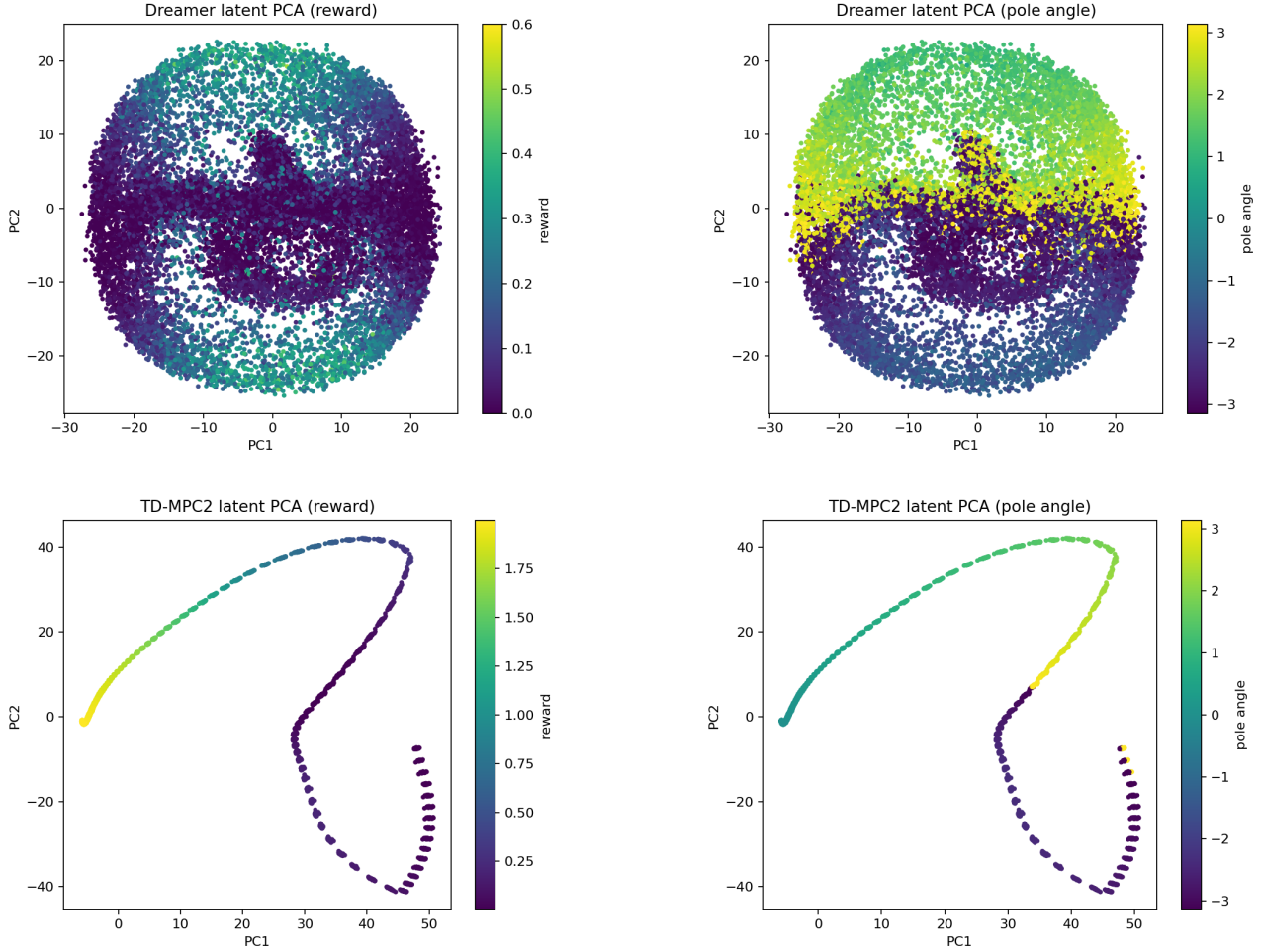


Figure 4. PCA visualization of the latent spaces on `cartpole-swingup`. Top row: Dreamer latents colored by reward and pole angle. Bottom row: TD-MPC2 latents colored by reward and pole angle. Dreamer yields a broader and more diffuse latent geometry, whereas TD-MPC2 yields a thinner, more structured manifold aligned with the task dynamics.

Table 2. Linear-probe results on `cartpole-swingup` latent representations.

Method	Target	R^2	MAE
Dreamer	Reward	0.7666	0.0341
Dreamer	Pole angle	0.4715	1.3647
Dreamer	Cart position	0.7993	0.2831
TD-MPC2	Reward	1.0000	0.0007
TD-MPC2	Pole angle	0.9429	0.0496
TD-MPC2	Cart position	1.0000	0.0000

explicitly around the control-relevant structure of the task. Dreamer still encodes substantial task information, especially for reward and cart position, but the representation is less linearly accessible and more diffuse in geometry.

A careful interpretation is important here. The latent comparison does not by itself prove that one representation is

universally better in all settings. Dreamer is optimized under an explicit predictive objective and was analyzed from replay states, which likely encourages a broader latent geometry. TD-MPC2 is optimized more directly for control and value estimation, and the extracted trajectories come from successful evaluation rollouts, which naturally favors a more compact task-aligned manifold. Nevertheless, the result strongly supports the conceptual distinction motivating this paper: explicit and implicit world models differ not only in learning curves, but also in the kind of internal representations they form.

5.4. Summary of the main benchmark

Taken together, the cartpole experiments yield three main findings. First, TD-MPC2 clearly outperforms Dreamer in mean control performance on the shared 150k horizon, reaching a final mean reward of 682.05 compared with

Dreamer’s 115.30. Second, Dreamer is more stable across seeds, whereas TD-MPC2 shows substantially larger run-to-run variability, as reflected in the much wider 95% confidence interval. Third, latent-space probing reveals a clear representational difference: TD-MPC2 learns a latent space that is much more linearly aligned with reward and key cartpole state variables, while Dreamer learns a broader and more distributed latent geometry.

These results support the broader claim of the paper: the difference between explicit and implicit world models is not limited to external performance alone. Rather, the two approaches also induce different optimization behavior across seeds and different internal organization of task-relevant information. On the main matched benchmark, the implicit TD-MPC2-style representation is more compact and control-aligned, whereas the explicit Dreamer-style representation is broader and less directly linear in the variables examined here.

5.5. Additional environment: walker-walk

To test whether the qualitative difference observed on the main matched benchmark extends beyond cartpole-swingup, we additionally evaluated both agents on `walker-walk` under a reduced-budget setting. In this extension, Dreamer was kept in the same lightweight state-based regime, while TD-MPC2 used `model.size=1` for computational efficiency. For fairness, the comparison is reported on the shared 95k horizon, since the Dreamer walker runs reached 96,096 environment steps and the plotting grid was aligned to 5k increments.

Figure 5 shows the three-seed mean and 95% confidence interval for both methods. The resulting learning curves again favor the implicit approach. At the shared 95k horizon, Dreamer reaches a final mean score of 38.64 with a 95% confidence-interval half-width of 4.01, whereas TD-MPC2 reaches a final mean evaluation reward of 972.46 with a 95% confidence-interval half-width of 5.36. The best mean value on the shared grid is 38.70 for Dreamer and 972.46 for TD-MPC2.

Unlike on cartpole-swingup, where TD-MPC2 achieved higher mean performance but also exhibited much larger variability across seeds, the walker-walk extension shows both a strong performance advantage and a relatively tight uncertainty band for TD-MPC2. Dreamer, in contrast, remains in a much lower-return regime throughout the run and shows no comparable improvement over the same budget. This extension therefore supports the same qualitative conclusion as the main benchmark: in the present lightweight setup, the TD-MPC2-style agent is substantially stronger in sample efficiency and final control performance than the Dreamer-style agent.

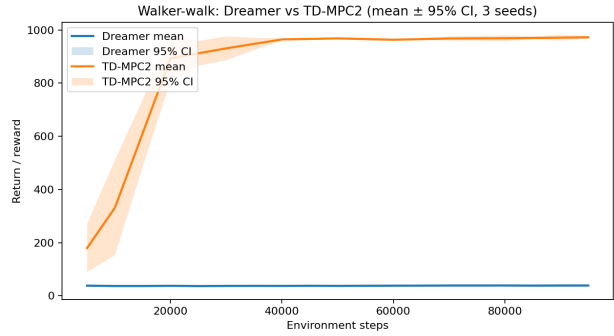


Figure 5. Walker-walk comparison over three seeds. Curves show the mean and shaded regions show 95% confidence intervals up to the shared 95k horizon. In this extension, TD-MPC2 uses `model.size=1`.

6. Conclusion

This paper compared two contrasting paradigms for world modeling in continuous control: an explicit Dreamer-style agent based on recurrent state-space modeling and imagined rollouts, and an implicit TD-MPC2-style agent based on latent consistency, value estimation, and latent-space planning. The study focused on low-dimensional state observations in order to isolate differences in latent modeling and control behavior rather than visual representation learning.

On the main matched benchmark, `cartpole-swingup`, the results showed a clear trade-off between performance, stability, and representation structure. TD-MPC2 achieved substantially stronger mean control performance than Dreamer on the shared comparison horizon, but also exhibited much larger variability across seeds. Dreamer, in contrast, produced lower returns on this task while appearing more stable across runs. This indicates that explicit and implicit world-model approaches should not be compared only in terms of peak performance, since their optimization behavior across seeds can differ substantially even in lightweight settings.

The latent-space analysis further showed that the two methods learn qualitatively different internal representations. Dreamer produced a broader and more diffuse latent geometry, whereas TD-MPC2 produced a thinner and more structured manifold. Linear probing showed that TD-MPC2 latent states were much more directly predictive of reward and key task-state variables, while Dreamer still encoded useful task information but in a less linearly accessible form. These results support the broader claim of the paper: explicit and implicit world models differ not only in external learning curves, but also in the kind of task-relevant structure they preserve internally.

Overall, the findings suggest a practical trade-off. Dreamer

provides a more distributed predictive latent space and more stable behavior across seeds in this setup, whereas TD-MPC2 yields stronger task performance and a more compact control-aligned representation, at the cost of greater seed sensitivity. Extending the same comparison protocol to additional environments such as `walker-walk` provides a natural next step for testing whether these trends persist under more complex dynamics.

A lower-budget extension on `walker-walk` showed the same qualitative pattern, with TD-MPC2 learning much faster and reaching a substantially stronger reward regime than Dreamer over the shared 95k horizon.

Preliminary extensions beyond the main matched benchmark indicate that the same evaluation protocol can be applied to more challenging locomotion tasks, allowing the comparison to be broadened beyond cartpole while preserving the same emphasis on performance, stability, and latent representation structure.

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